MILITARY SPECIFICATION

Fittings, Automotive Tube, Brass, Inverted Flare

(With Steel Nut)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for tube fittings of the inverted flare type (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-S-637 - Steel Bar, Carbon, Cold Finished (Standard Quality Free Machining).
QQ-W-403 - Wire, Steel, Carbon and Alloy (For Cold Heading and Cold Forging).
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements for.
WW-T-775 - Tube, Copper, Seamless (For Refrigeration and General Use).

Military

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein); Packaging of.

STANDARDS

Federal

Federal Test Method Std. No. 151 - Metals: Test Methods.

FSC 4730

Source: https://assist.dla.mil -- Downloaded: 2026-
Check the source to verify that this is the current ver

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-129	-	Marking for Shipment and Storage.
MS51874	-	Inverted Nut, Tube Coupling, Flared Type (Design Standard).
MS51875	-	Adapter, Straight, Pipe to Tube, Male Pipe End, Inverted Flare Type.
MS51876	-	Adapter, Straight, Pipe to Tube, Male Pipe End, Inverted Flare Type.
MS51877	-	Nipple, Tube, Inverted Flare Type.
MS51878	-	Elbow, Pipe to Tube, 45°, Male Pipe End, Inverted Flare Type.
MS51879	-	Elbow, Pipe to Tube, 90°, Male Pipe End, Inverted Flare Type.
MS51880	-	Tee, Tube, Inverted Flare Type.
MS51881	-	Tee, Pipe to Tube, Male Pipe on Run, Inverted Flare Type.
MS51882	-	Tee, Pipe to Tube, Female Pipe on Branch, Inverted Flare Type.
MS51883	-	Tee, Pipe to Tube, Male Pipe on Branch, Inverted Flare Type.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402.)

Society of Automotive Engineers (SAE)

SAE J461 - Wrought and Cast Copper Alloys
SAE J512 - Automotive Tube Fittings

(Application for copies should be addressed to the Society of Automotive Engineers (SAE) Standards Committee, 485 Lexington Avenue, New York, N.Y., 10017.)

Source: https://assist.dla.mil -- Downloaded: 2026
Check the source to verify that this is the current ve

American Trucking Association, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N.W., Washington, D.C., 20036.)

Uniform Classification Committee Agency

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois, 60606.)

(Technical society and technical association specification and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Material. Fitting bodies shall be made of brass in accordance with the following:

a. For bar or extruded stock use QQ-B-626, Alloy CA360, temper half-hard or SAE J461, Alloy SAE CA360, half-hard.

b. For forging use QQ-B-626, Alloy CA377 or SAE J461, Alloy SAE CA377.

Nuts shall be made of steel in accordance with QQ-S-637, 12L14 or QQ-W-405, cold headed stock or commercial equivalent.

3.2 Finish. Brass bodies shall be uncoated. Steel nuts shall be cadmium plated in accordance with QQ-P-416, type II, class 3 or zinc coated in accordance with QQ-Z-325, type II, class 3. (It is preferred that zinc coating be used instead of cadmium plating.)

3.3 Design.

3.3.1 Construction. Fittings, when properly assembled using copper seamless tubing WW-T-775, shall insure a leakproof full flow connection up to 2100 pounds per square in. (PSI) working pressure depending on the size and wall thickness of the tubing used (see Table I). This is the established recommended pressure based on a safety factor of four to one.

Source: https://assist.dla.mil -- Downloaded: 2026-10
Check the source to verify that this is the current versi

Table I

Copper Tubing in Accordance with WW-T-775

Tube OD	Wall Thickness	Working Pressure	Proof Pressure
1/8	.030	2100	4200
3/16	.030	1400	2800
1/4	.030	1050	2100
5/16	.032	900	1800
3/8	.032	750	1500
7/16	.032	650	1300
1/2	.032	575	1150
5/8	.035	500	1000
3/4	.035	400	800

3.3.2 Dimensions. Fitting dimensions shall be in accordance with MS51874 thru MS51883 and, if not specified, in accordance with the Society of Automotive Engineers Standard SAE J512.

3.3.3 Shape and form restrictions. Abrupt reductions of sections shall be avoided. Small external sections adjoining relatively heavy body sections shall be shaded into the heavier sections by means of ample fillets. Sharp corners, inadequate fillets, excessive undercuts, or excessive grooves at the junction of such small sections with large sections of fittings shall be considered cause for rejection.

3.3.4 Passages. Where passages in straight fittings are machined from opposite ends, the offset at the meeting point shall not exceed 0.015 inch. The cross sectional area at the junction of passages in angle fittings shall not be less than that of the smallest passage. Where the passage is specified as a maximum, or as tap drill diameter or less, the minimum shall be no less than the minimum diameter of the smallest passage in the fitting.

3.3.5 Wall thickness. Unless otherwise designated, the wall thickness at any point on fittings shall not be less than the thickness established in SAE J512.

3.3.6 Straight threads. Straight threads shall conform to the Unified Screw Thread Standard, Class 2A and 2B, in accordance with the Screw Thread Standard for Federal Services, Handbook H28, except that Class 3B minor diameter limits shall apply to copper alloy fittings. Class 2B minor diameter limits shall apply to steel fittings.

3.3.7 Taper pipe threads. Taper pipe threads shall conform to the American National Standard Pipe Threads (Dry Seal-NPTF) in accordance with Handbook H28.

Source: https://assist.dla.mil -- Downloaded: 2026-
Check the source to verify that this is the current ve

3.4 Marking. Unless otherwise specified, all fittings shall be marked with manufacturer's name or trade mark. The marking shall be applied in a location not detrimental to the fitting or to the corrosion protection on the nut.

3.5 Workmanship. Workmanship shall conform to the best commercial practice to produce high quality fittings. Fittings shall be free from all hanging burrs, loose scale, and slivers which might affect their serviceability. All sealing surfaces must be smooth except that annular tool marks up to 100 microinch maximum shall be permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms. Inspection terms used in this specification are defined in MIL-STD-109.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Lot. A lot shall consist of all fittings of the same material, type, style, finish, heat treat and size, produced by the same manufacturer under essentially the same conditions, and submitted for acceptance at one time.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of fittings and nuts shall be taken from each lot in accordance with MIL-STD-105, Inspection Level S4. The Acceptable Quality Level (AQL) shall be as specified in Table II.

4.4 Inspection.

4.4.1 Lot acceptance inspection for fittings and nuts. Fittings and nuts selected in accordance with 4.3.1 shall be visually and dimensionally inspected to verify compliance with the requirements of this specification. The classification of defects shall be as specified in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-1
Check the source to verify that this is the current vers

Table II - Classification of Defects

Categories	Defects	Inspection Method
Major	AQL = 2.5 percent defective	
101	Fittings and nuts, not type specified	SIE*
102	Sealing surface, not as specified (3.5)	SIE*
103	Dimensions (3.3.2)	SIE*
104	Threads, not as specified (3.3.6 and 3.3.7)	SIE*
105	Evidence of use of unauthorized materials	SIE*
106	Shape and form reductions abrupt; small external sections not shaded into heavy body sections by fillets; fittings with sharp corners; inadequate fillets; excessive undercuts and grooves (see 3.3.3)	Visual
107	Bores drilled from each end exceed specified tolerance at drill meeting point; size of fluid passage less than specified; fluid passage of angle fittings, not as specified (3.3.4)	SIE*
108	Wall thickness of fittings, less than dimension and tolerance specified on applicable documents (3.3.5)	SIE*
109	Protective coating, not as specified (3.2)	Visual
Minor	AQL = 4.0 percent defective	
201	Marking, not located as specified, not permanent, not legible, data not complete (3.4)	Visual

* Standard Inspection Equipment

4.5 Lot acceptance test. Inverted flared fittings shall be tested as specified in 4.5.1 thru 4.5.2.2. Fittings that fail to pass these tests shall be considered defective and lot represented by the fittings shall be rejected.

4.5.1 Chemical analysis. Unless otherwise specified (see 6.2), certification showing conformance with the applicable material specification shall be available to the Government. When specified (see 6.2), chemical analysis shall be in accordance with Method 111.2 of Federal Test Method Standard No. 151.

4.5.2 Performance test.

4.5.2.1 Tubing used in test. Copper tubing shall be in accordance with WW-T-775.

Source: https://assist.dla.mil -- Downloaded: 2026-10
Check the source to verify that this is the current versi

4.5.2.2 Test. The fittings selected in accordance with 4.3.1 shall be assembled with properly flared tubing, and the tapered threads plugged and capped as applicable. The tube length shall be at least eight times the tube diameter. The steel nut(s) shall be hand tightened to the assembly body, then with a proper wrench tighten until a solid feeling is encountered. From that point, apply one-sixth turn. The assembly shall then be tested to the applicable proof pressure given in Table I. Under this pressure, there shall be no leakage or distortion of the assembly. The assembly shall then be disassembled and the same procedure repeated until the fitting(s) have been assembled and pressure applied eight successive times.

4.6 Inspection and preparation for delivery. The packaging, packing and marking shall be inspected for compliance with Section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Fittings shall be packaged in quantities as specified in accordance with the Level A packaging requirements of MIL-V-3.

5.1.2 Level C. Fittings shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. The fittings shall be packed in accordance with the applicable Level A requirements of MIL-V-3.

5.2.2 Level B. The fittings shall be packed in accordance with the applicable Level B requirements of MIL-V-3.

5.2.3 Level C. The fittings shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the contract or order, interior container and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-
Check the source to verify that this is the current ver

6. NOTES

6.1 Intended use. Fittings covered by this specification are intended for use with copper, aluminum, and steel tubing for general applications in the automotive field.

6.2 Ordering data. Procurement should specify the following:

a. Title, number and date of this specification.
b. MS part number - use MS number plus its dash number (2.1).
c. Material (3.1).
d. Protective finish (3.2).
e. Chemical analysis test when specified (see 4.5.1).
f. Selection of applicable level of packaging and packing required (5.1 and 5.2).

Custodian:
Army - MC
Navy - OS
Air Force - 82

Preparing Activity:
Army - MC

Project No. 4730-0362

Review Activities:
Army - AT
Navy - None
Air Force - None
DSA - CS

User Activities:
Army - ME
Navy - MC
Air Force - None

Source: https://assist.dla.mil -- Downloaded: 2026-10
Check the source to verify that this is the current versio